

MATA37H3 — Week 12

Power Series II: Taylor Polynomials

60 minutes (Pages 1–5, self-paced) • OpenStax *Calculus Volume 2*, §6.3 (Taylor and Maclaurin Series) — Definition of the n th Taylor polynomial, Theorem 6.5 (Uniqueness of Power Series).
MAT137 Unit 14 slides (14.4–14.6: Taylor polynomials of a polynomial from two equivalent definitions; reconstructing a polynomial from its derivative values; the “Competition” task showing Taylor coefficients are basis-independent) — MAT137 is the University of Toronto’s official exclusion-equivalent of MATA37H3.

DEFINITION The n th Taylor Polynomial

If f has n derivatives at $x = a$, the **n th Taylor polynomial for f at a** is

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

When $a = 0$, p_n is called the **n th Maclaurin polynomial**.

KEY FACT Theorem 6.5: Uniqueness of Power Series

If f can be written as **any** polynomial (or power series) in $(x-a)$, its Taylor polynomial coefficients are exactly the coefficients of that expansion — you don’t have to differentiate to find them. Equivalently: at most one degree- $\leq n$ polynomial can match $f(a), f'(a), \dots, f^{(n)}(a)$ simultaneously, so any correct method for finding p_n (differentiating and plugging in, or recognizing f ’s expansion directly) must produce the **same** answer.

KEY FACT Taylor Polynomials of a Polynomial (No Differentiation Needed)

If f is already a polynomial of degree d and $n \geq d$, then $p_n(x) = f(x)$ **exactly** — the Taylor polynomial reproduces f with zero remainder, regardless of the center a (only the *packaging* into powers of $(x-a)$ changes). If $n < d$, p_n keeps exactly the terms of degree $\leq n$ from f ’s expansion in powers of $(x-a)$ and drops the rest — including any terms that happen to be zero, which must still be accounted for when reasoning through the pattern.

KEY FACT Reconstructing a Polynomial from Derivative Data

Given $f(a), f'(a), f''(a), \dots, f^{(n)}(a)$, the coefficient formula is **forced**, not a choice: there is exactly one degree- $\leq n$ polynomial with those derivative values at a (Theorem 6.5, applied to the finite case), so simply plugging the given values into the coefficient formula produces the unique correct answer.

Worked Examples

EXAMPLE 1

Taylor polynomials of a polynomial from the coefficient formula, matched against direct algebra

For $f(x) = x^3$, find $p_3(x)$ and $p_2(x)$ at $a = 0$, two ways.

Solution. Coefficient formula: $f(0) = 0$, $f'(x) = 3x^2 \Rightarrow f'(0) = 0$, $f''(x) = 6x \Rightarrow f''(0) = 0$, $f'''(x) = 6 \Rightarrow f'''(0) = 6$. So $p_3(x) = 0 + 0 \cdot x + \frac{0}{2!}x^2 + \frac{6}{3!}x^3 = x^3$. **Direct check**

(Uniqueness of Power Series): $f(x) = x^3$ is *already* a polynomial in x of degree 3, so its own Taylor polynomial of degree ≥ 3 must just be itself — matches. For p_2 : truncate the degree-3 expansion at degree 2, i.e. drop the x^3 term: $p_2(x) = 0$ (the zero polynomial), since x^3 contributes nothing to degrees 0, 1, 2. (*Misconception flag: $p_2(x)$ is 0, not “undefined” or “ x^3 with the x^3 erased differently” — the n th Taylor polynomial keeps exactly the terms of degree $\leq n$ from the true expansion and nothing else.*) ■

EXAMPLE 2

same f , centered at $a = 1$

For $f(x) = x^3$, find $p_3(x)$ centered at $a = 1$ (write it in powers of $(x - 1)$).

Solution. $f(1) = 1$, $f'(1) = 3$, $f''(1) = 6$, $f'''(1) = 6$. So

$$p_3(x) = 1 + 3(x - 1) + \frac{6}{2!}(x - 1)^2 + \frac{6}{3!}(x - 1)^3 = 1 + 3(x - 1) + 3(x - 1)^2 + (x - 1)^3.$$

Check by substitution: let $u = x - 1$, so $x = u + 1$ and $x^3 = (u + 1)^3 = u^3 + 3u^2 + 3u + 1$ — matches term-for-term with $u = (x - 1)$. Since f is a cubic, $p_3 = f$ **exactly** (no approximation, no remainder) regardless of the center — only the *packaging* into powers of $(x - a)$ changes. ■

EXAMPLE 3

reconstructing a polynomial from derivative data — building toward the coefficient formula

A polynomial P of degree ≤ 3 satisfies $P(0) = 5$, $P'(0) = -2$, $P''(0) = 0$, $P'''(0) = 12$. Find $P(x)$, and explain why it is the *only* such polynomial of degree ≤ 3 .

Solution. By the coefficient formula (which is forced, not a choice, once we require $P = p_3$ for itself): $P(x) = 5 + (-2)x + \frac{0}{2!}x^2 + \frac{12}{3!}x^3 = 5 - 2x + 2x^3$. **Uniqueness:** any degree- ≤ 3 polynomial is completely determined by its value and first three derivatives at a single point, because matching $Q(0) = P(0)$, $Q'(0) = P'(0)$, $Q''(0) = P''(0)$, $Q'''(0) = P'''(0)$ for two degree- ≤ 3 polynomials forces $Q - P$ to be a degree- ≤ 3 polynomial with a quadruple root at 0 in the derivative sense — i.e. $Q - P \equiv 0$ (Theorem 6.5, Uniqueness of Power Series, applied to the finite case). ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

Taylor polynomials of a polynomial (no differentiation needed — use uniqueness)

1. For $f(x) = 2x^3 - x^2 + 5$, write down $p_3(x)$, $p_2(x)$, and $p_1(x)$ at $a = 0$ directly from the given expansion, with no derivative computations.

HINT Read the coefficients directly off the given polynomial $f(x) = 2x^3 - x^2 + 5$ — no derivatives are needed thanks to uniqueness.

CAUTION — before you solve Don't differentiate from scratch to find p_2 or p_1 — just drop the higher-power terms from f itself; that's what uniqueness guarantees.

Your working:

2. For $f(x) = x^3$, find $p_3(x)$ centered at $a = -1$ (powers of $(x + 1)$), and verify it by substituting $u = x + 1$ as in Example 2.

HINT Substitute $u = x + 1$ (so $x = u - 1$), expand $(u - 1)^3$, then translate back to powers of $(x + 1)$.

CAUTION — before you solve Watch the signs when expanding $(u - 1)^3$ — the odd-power terms alternate sign, and it's easy to drop a minus sign.

Your working:

3. A polynomial Q of degree ≤ 4 satisfies $Q(2) = 1$, $Q'(2) = 0$, $Q''(2) = -6$, $Q'''(2) = 0$, $Q^{(4)}(2) = 24$. Find $Q(x)$ in powers of $(x - 2)$.

HINT Plug the given derivative values directly into the Taylor coefficient formula $\frac{f^{(k)}(2)}{k!}(x-2)^k$ for each k from 0 to 4.

CAUTION — before you solve Don't skip the zero-derivative terms silently — include $0 \cdot (x-2)$ and $0 \cdot (x-2)^3$ in your reasoning so you don't mismatch a power with the wrong given value.

Your working:

Taylor polynomials of a non-polynomial function (differentiation required)

4. Find $p_3(x)$ (the 3rd Maclaurin polynomial) for $f(x) = e^x$, using $f(0) = f'(0) = f''(0) = f'''(0) = 1$.

HINT Compute $f(0), f'(0), f''(0), f'''(0)$ for e^x (all equal 1) and plug into the coefficient formula.

CAUTION — before you solve Don't forget to divide each derivative value by the matching factorial — the coefficients are $\frac{f^{(k)}(0)}{k!}$, not the raw derivative values.

Your working:

5. Find $p_2(x)$ for $f(x) = \cos x$ centered at $a = 0$, using $f(0) = 1, f'(0) = 0, f''(0) =$

−1.

HINT Compute $f(0)$, $f'(0)$, $f''(0)$ for $\cos x$ and plug into the coefficient formula through degree 2.

CAUTION — before you solve Track the sign of $f''(0) = -1$ carefully — a dropped negative sign flips the concavity of the resulting quadratic approximation.

Your working:

Reasoning / “why does this work”

6. Two students each write $f(x) = x^4$ as a Taylor expansion at $a = 0$ using totally different-looking methods (one differentiates four times and plugs into the coefficient formula; the other multiplies out $f(x) = x \cdot x \cdot x \cdot x$ directly). Explain, citing Theorem 6.5, why they are guaranteed to get the same $p_4(x)$ — and give it.

HINT Think about what Theorem 6.5 (Uniqueness of Power Series) guarantees about how many degree- $\leq n$ polynomials can match f and its first n derivatives at a point.

CAUTION — before you solve Two different-looking correct methods can’t give two different Taylor polynomials — if your answer disagrees with someone else’s, at least one computation has an arithmetic error, not a ‘different but valid’ answer.

Your working:

7. True or false, with justification: “If f is a polynomial of degree 7, then its Taylor polynomial $p_{10}(x)$ at any center a equals $f(x)$ exactly.” (Careful with the direction of the inequality 10 vs. 7.)

HINT Compare the polynomial’s actual degree to 10, and think about what happens to the higher-order terms of the Taylor coefficient formula when f is already a polynomial.

CAUTION — before you solve Don’t reflexively assume higher-degree Taylor polynomials are only ever ‘better approximations’ — once $n \geq \deg f$, p_n equals f exactly, with zero remainder.

Your working:

8. Explain why, for a polynomial f of degree d , the Taylor polynomial $p_n(x)$ for $n < d$ is generally **not** the same as simply setting $x^{n+1}, x^{n+2}, \dots$ to zero in f ’s expansion **around a nonzero center** a — i.e., why the truncation must happen in powers of $(x - a)$, not in powers of x .

HINT Think about what happens to x^k itself when re-expanded in powers of $(x - a)$ via the binomial theorem, for a nonzero center a .

CAUTION — before you solve Zeroing out high powers of x and zeroing out high powers of $(x - a)$ are only the same operation when $a = 0$ — for a nonzero center they redistribute information differently.

Your working:
